

Data Communication

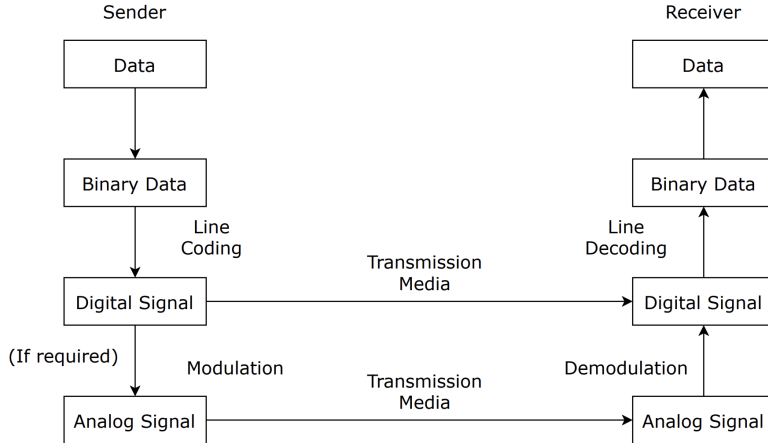
Content 04: Digital Transmission

Abdullah Ibne Masud Mahi (AIMM)

Department of Computer Science and Engineering
United International University



Data Transmission Pipeline



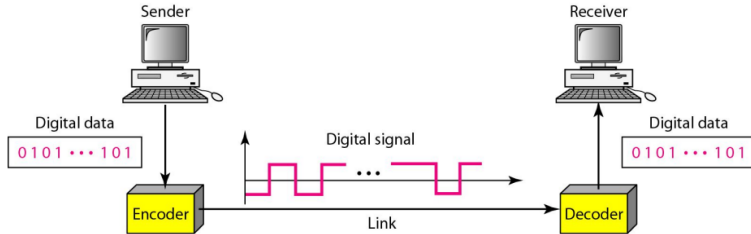
What is Digital Transmission?

- Transmission of digital signals over a medium.
- Uses discrete values instead of continuous signals.
- Examples: Ethernet, Wi-Fi, Fiber-optic communication.



Digital-Digital Conversion

- Process of converting digital data into a digital signal.
- Uses line coding, block coding and scrambling techniques.
- Ensures proper transmission and synchronization.



What is Line Coding?

- Encoding digital data into digital signals.
- Ensures compatibility with transmission medium.
- Different schemes affect bandwidth, synchronization, and error detection.

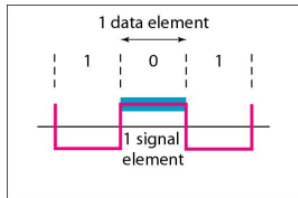


Signal Element and Data Element

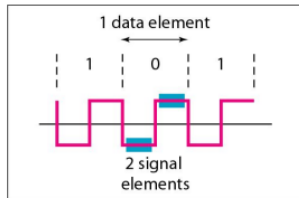
- **Data Element:** Smallest unit of data (bit, character).
- **Signal Element:** Smallest unit of a signal that represents data.
- **Relation:** Data elements are mapped to signal elements for transmission.



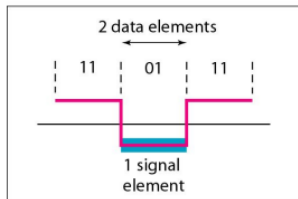
Signal Element and Data Element



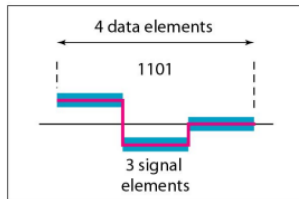
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements ($r = \frac{1}{2}$)



c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements ($r = \frac{4}{3}$)

Data Rate and Bit Rate Relation

- **Bit Rate:** Number of bits transmitted per second (bps).
- **Baud Rate:** Number of signal elements per second (baud).

$$S = c * N * \frac{1}{r}$$

Where-

- S = Signal Rate (baud)
- c = Case Factor (Efficiency for a particular scheme)
- N = Data Rate (bps)
- r = Ratio ($\frac{Data}{Signal}$)



Data Rate and Bit Rate Relation

A signal is carrying data in which one data element is encoded as one signal element ($r = 1$). If the bit rate is 100 kbps, what is the average value of the baud rate if c is between 0 and 1?

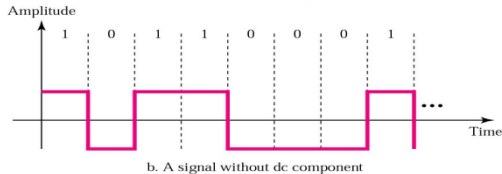
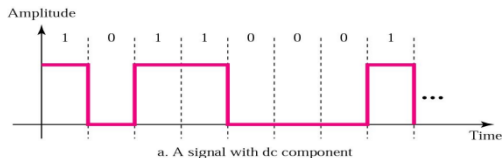
We assume that the average value of c is $1/2$. The baud rate is then-

$$S = c * N * \frac{1}{r} = \frac{1}{2} * 100 * \frac{1}{1} = 50 \text{ kbaud}$$



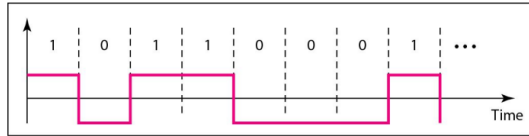
Problems of Line Coding: DC Component

- The DC component problem occurs when a signal has a constant voltage level, which can cause signal distortion or loss in certain transmission systems.
- DC component causes problems in AC-coupled transmission.
- Solution: Use bipolar encoding (e.g., AMI) to remove the DC component.

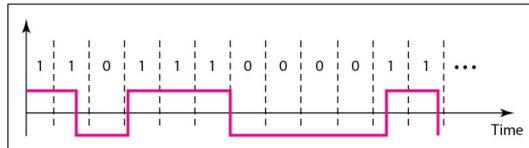


Problems of Line Coding: Self-Synchronization

- Long sequences of identical bits cause loss of synchronization.
- Self-clocking codes (e.g., Manchester encoding) solve this issue.

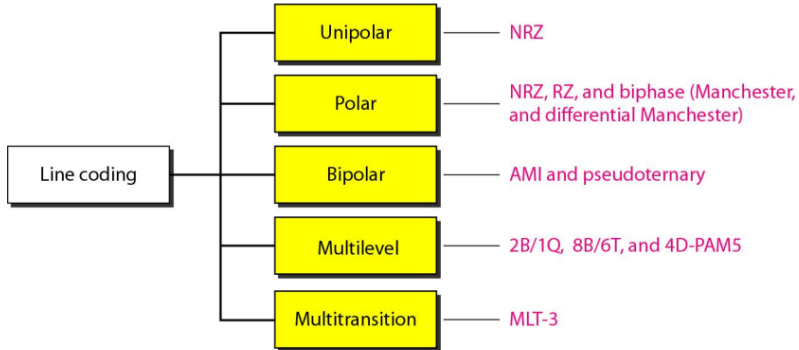


a. Sent



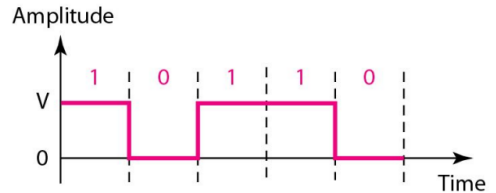
b. Received

Classification of Line Coding



Unipolar NRZ

- One voltage level; $1=V$, $0=0V$
- Simple, easy to implement
- DC component, no synchronization



Polar NRZ-L and NRZ-I

NRZ-L

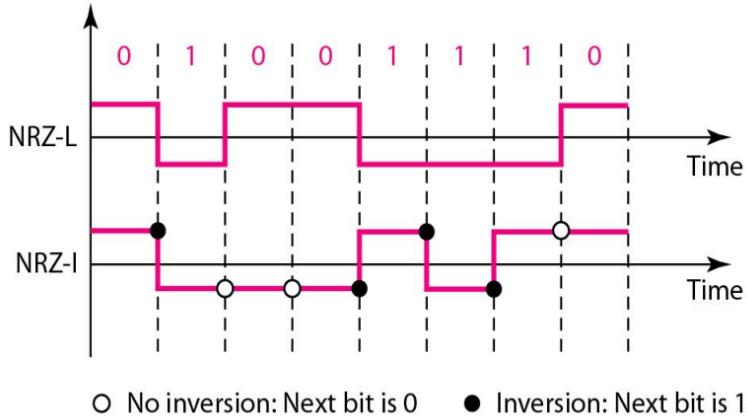
- 1 = Negative, 0 = Positive voltage
- Reduces DC
- No synchronization

NRZ-I

- Change at 1, No change at 0
- Better error detection
- Still no inherent synchronization

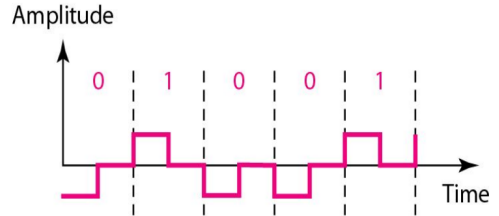


Polar NRZ-L and NRZ-I



Polar RZ

- 1 = Positive to Zero
- 0 = Negative to Zero
- Self-clocking
- Requires more bandwidth



Manchester and Differential Manchester

Manchester

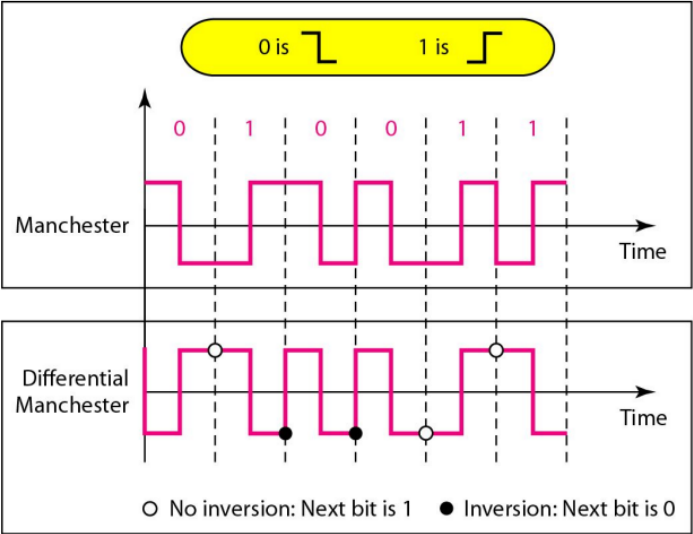
- 1=Transition Low-High,
0=Transition High-Low
- Self-synchronizing
- Requires double bandwidth

Differential Manchester

- Transition at 0, No transition at 1
- Better noise immunity
- More complex decoding



Manchester and Differential Manchester



AMI and Pseudoternary

AMI

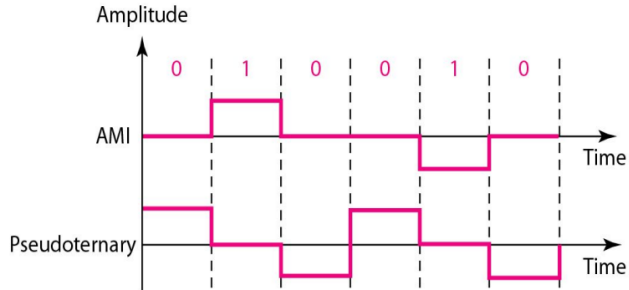
- 1 = Alternates between $+V/-V$
- 0 = 0 V
- No DC component
- Requires synchronization

Pseudoternary

- 0 = Alternates $+V/-V$, 1 = 0V
- No DC component
- Less common usage



AMI and Pseudoternary



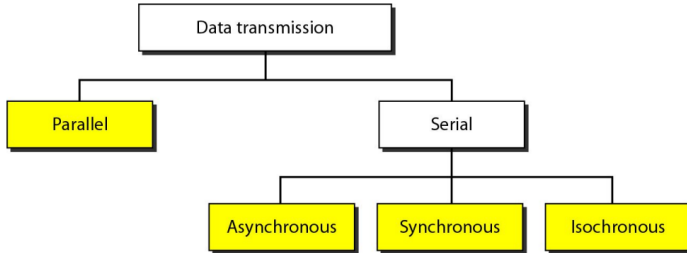
Transmission Modes

- Data transmission is the process of sending data from one device to another over a communication medium, such as wires or wireless signals.
- It can happen in different ways, like sending multiple bits at once (parallel) or one bit at a time (serial).



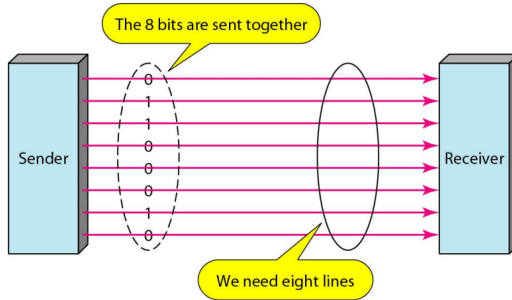
Classification of Transmission Modes

- Parallel Transmission - Multiple bits sent at a time.
- Serial Transmission - Bits sent one after another.



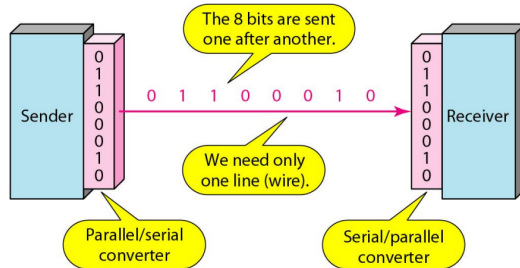
Parallel Transmission

- Data is sent multiple bits at a time using multiple channels.
- Faster than serial transmission but limited to short distances.
- Used in ****computer buses, RAM, and printers****.



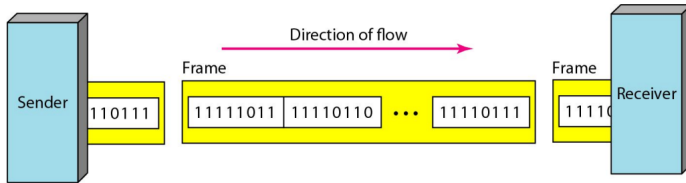
Serial Transmission

- Data is sent one bit at a time over a single channel.
- Used for long-distance communication (e.g., USB, Ethernet).
- Slower than parallel but requires fewer wires.



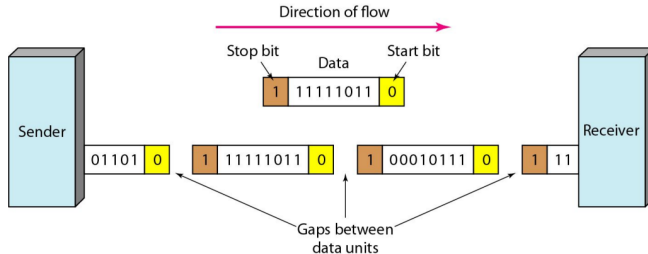
Synchronous Transmission

- Data is sent in a continuous stream with a clock signal.
- No start/stop bits; relies on synchronization between sender and receiver.
- Used in high-speed networks (Ethernet, SDH, ISDN, etc.).



Asynchronous Transmission

- Data is sent in small independent units (frames).
- Each unit has start and stop bits for synchronization.
- Slower than synchronous but cost-effective and simple.
- Used in keyboards, RS-232 serial communication.



Practice Problems

- Theoretical:
Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 04 - 2, 5, 11
- Mathematical:
Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 04 - 15, 16, 17, 18



References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 04 - 4.1, 4.3



Thank You

